

The Ghoshian Dual Parallel State–Attention Architecture

A Unified Cognitive, Economic, and
Computational Framework for
Adaptive Foundation Models

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Abstract

Modern foundation models are dominated by Transformer architectures, while recent State Space Models (SSMs) offer compelling advantages in memory efficiency and long-context processing. Each paradigm possesses complementary strengths and weaknesses. Simultaneously, Ghoshian Learning introduces a meta-learning framework in which learning rates and optimization dynamics are themselves learned through breadth and depth signals.

This paper develops the Ghoshian Dual Parallel State–Attention Architecture (G-DPSA), a unified framework combining Transformers, State Space Models, Mixture-of-Experts systems, hierarchical memory, reinforcement learning, and Ghoshian Learning into a single adaptive cognitive architecture.

The resulting system behaves as a self-regulating computational economy in which reasoning, memory, learning, exploration, exploitation, attention allocation, and resource management are jointly optimized. The framework is analyzed from perspectives including mathematics, computer science, machine learning, statistics, economics, finance, psychology, philosophy, political science, international relations, warfare economics, and welfare economics.

The paper ends with “The End”

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1 Introduction

Large-scale neural systems currently face a fundamental tradeoff.

Transformers provide exceptional reasoning capabilities but suffer from quadratic attention complexity:

$$O(n^2).$$

State Space Models provide efficient long-range memory:

$$O(n),$$

but often sacrifice global reasoning ability.

Simultaneously, optimization methods remain largely dependent upon externally specified hyperparameters.

Ghoshian Learning introduces a radically different paradigm in which learning rates themselves become learnable objects.

This paper proposes a synthesis:

1. Transformer reasoning,
2. State Space memory,
3. Ghoshian executive control,
4. Hierarchical memory,
5. Sparse expert specialization,
6. Reinforcement-based compute allocation.

The resulting architecture resembles a computational civilization rather than a traditional neural network.

2 Background

2.1 Transformers

Transformers compute

$$Q = XW_Q,$$

$$K = XW_K,$$

$$V = XW_V.$$

Attention is

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V.$$

Advantages include:

- strong reasoning,
- in-context learning,
- symbolic manipulation,
- multimodal integration.

Disadvantages include:

- quadratic complexity,
- large memory requirements,
- expensive inference.

2.2 State Space Models

State evolution follows

$$s_t = A(x_t)s_{t-1} + B(x_t)x_t.$$

Output generation follows

$$h_t = C(x_t)s_t.$$

Advantages include:

- linear scaling,
- persistent memory,
- efficient inference.

Disadvantages include:

- weaker symbolic reasoning,
- weaker retrieval mechanisms,
- potential memory dilution.

2.3 Ghoshian Learning

Ghoshian Learning treats learning rates as learnable objects:

$$\eta_t = R_\phi(s_t).$$

Learning is governed by two quantities:

- Breadth:

$$b_t = \frac{\text{Var}(g_t) + \epsilon}{\|E[g_t]\|^2 + \epsilon}$$

- Depth:

$$d_t = \exp\left(-\gamma \frac{\Delta \tilde{L}_t}{\tilde{L}_{t-1} + \epsilon}\right)$$

Breadth promotes exploration.

Depth promotes convergence.

3 The Ghoshian Dual Parallel State–Attention Architecture

3.1 Overview

The architecture consists of three interacting systems:

1. Transformer subsystem,
2. State Space subsystem,
3. Ghoshian Executive Controller.

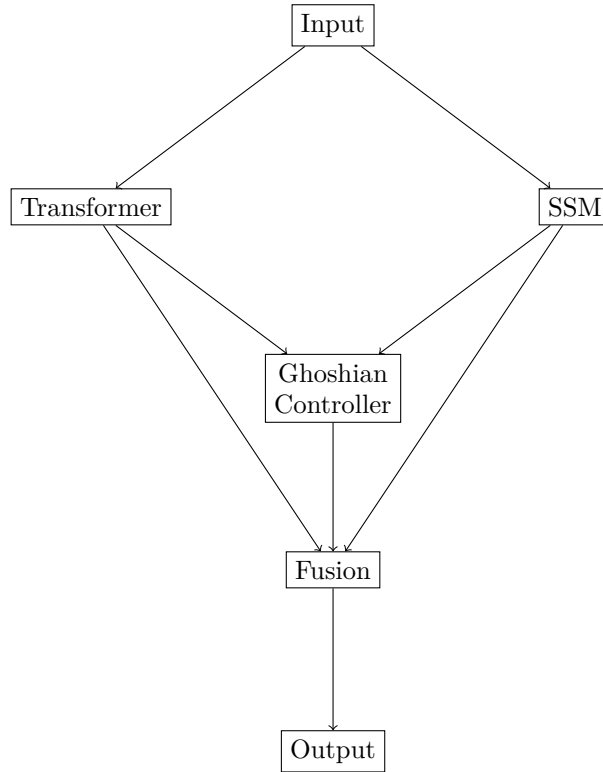


Figure 1: The Ghoshian Dual Parallel State–Attention Architecture.

4 Mathematical Formulation

4.1 Transformer Representation

Let

$$h_T = T_{\theta_T}(x).$$

4.2 State Space Representation

Let

$$h_S = S_{\theta_S}(x).$$

4.3 Fusion

Fusion is governed by a learnable gate:

$$g_t = R_\phi(B_t, D_t).$$

The final representation becomes

$$h_t = g_t h_T + (1 - g_t) h_S.$$

5 Breadth and Depth Generalization

5.1 Transformer Breadth

Define

$$b_t^T = \frac{\text{Var}(A_t)}{\|E[A_t]\|^2 + \epsilon}.$$

5.2 SSM Breadth

Define

$$b_t^S = \frac{\text{Var}(s_t)}{\|E[s_t]\|^2 + \epsilon}.$$

5.3 Global Breadth

$$B_t = w_T b_t^T + w_S b_t^S.$$

5.4 Global Depth

$$D_t = \lambda_T d_t^T + \lambda_S d_t^S.$$

6 Adaptive Control Vector

The controller outputs

$$u_t = (\eta_t^T, \eta_t^S, \beta_t, g_t, a_t, m_t, r_t).$$

where

Variable	Meaning
η_t^T	Transformer learning rate
η_t^S	SSM learning rate
β_t	Momentum coefficient
g_t	Fusion gate
a_t	Attention budget
m_t	Memory budget
r_t	Expert routing policy

7 Hierarchical Ghoshian Learning

Layer-specific learning rates are generated:

$$\eta_{T,\ell} = R_\phi^\ell(s_t).$$

Similarly,

$$\eta_{S,k} = R_\phi^k(s_t).$$

Updates become

$$\theta_{T,\ell}^{t+1} = \theta_{T,\ell}^t - \eta_{T,\ell} g_{T,\ell},$$

$$\theta_{S,k}^{t+1} = \theta_{S,k}^t - \eta_{S,k} g_{S,k}.$$

8 Ghoshian Momentum

The controller jointly learns

$$(\eta_t, \beta_t).$$

Momentum evolves as

$$m_{t+1} = \beta_t m_t + (1 - \beta_t) g_t.$$

Parameter updates become

$$\theta_{t+1} = \theta_t - \eta_t m_{t+1}.$$

9 Ghoshian Reinforcement Learning

The architecture learns a policy

$$\pi_\phi.$$

Reward is

$$R_t = \alpha Q_t - \beta C_t + \gamma M_t.$$

where

- Q_t = performance,
- C_t = computational cost,
- M_t = memory efficiency.

The optimization problem becomes

$$\max_{\phi} E \left[\sum_t R_t \right].$$

10 Mixture of Experts

Let

$$E_1, \dots, E_n$$

denote expert modules.
Routing probabilities are

$$r_t = \text{softmax}(W_r h_t).$$

Experts may include:

- mathematics,
- code,
- language,
- audio,
- vision,
- long-term memory.

11 Theoretical Analysis

Theorem 1 (Adaptive Stability). *Suppose*

$$0 < \eta_{\min} \leq \eta_t \leq \eta_{\max} < \frac{2}{L}.$$

*Suppose breadth and depth signals are Lipschitz continuous.
Then the Ghoshian controller preserves SGD stability.*

Proof. The update is

$$\theta_{t+1} = \theta_t - \eta_t g_t.$$

By smoothness,

$$L(\theta_{t+1}) \leq L(\theta_t) - \eta_t \left(1 - \frac{L\eta_t}{2}\right) \|g_t\|^2.$$

Since

$$\eta_t < \frac{2}{L},$$

the descent coefficient remains positive.
Therefore expected loss decreases monotonically.

□

Theorem 2 (Compute-Efficiency Dominance). *Suppose the controller activates attention with probability p .*

Expected computational complexity becomes

$$E[C] = p O(n^2) + (1 - p) O(n).$$

For sufficiently small p ,

$$E[C] < O(n^2).$$

Proof. Direct substitution yields

$$E[C] = pn^2 + (1 - p)n.$$

For

$$0 < p < 1,$$

the expression lies strictly below quadratic growth. □

12 Economic Interpretation

The architecture behaves as a self-regulating economy.

Economic Concept	Neural Analogue
Capital	Parameters
Labor	Compute
Interest Rate	Learning Rate
Central Bank	Controller
Diversification	Breadth
Compounding	Depth
Investment	Attention Allocation
Budgeting	Memory Allocation

Breadth resembles portfolio diversification.

Depth resembles compounded returns.

Learning rates behave as endogenous interest rates.

13 Psychological Interpretation

The architecture naturally decomposes into:

Psychological Concept	Module
Working Memory	Transformer
Long-Term Memory	SSM
Executive Function	Controller
Attention	Dynamic Budget
Skill Learning	Experts
Metacognition	RL Controller

14 Philosophical Interpretation

The architecture addresses classical questions concerning:

1. Memory versus reasoning,
2. Exploration versus exploitation,
3. Knowledge versus inference,
4. Agency versus determinism.

The controller functions as a computational analogue of practical reason.

15 Political Science and International Relations

The architecture resembles a federal political system.

Transformer experts function as specialized ministries.

The controller functions as a central government.

Breadth measures political diversity.

Depth measures institutional stability.

Federated Ghoshian Learning resembles multilateral governance systems in international relations.

16 Welfare Economics

Define welfare

$$W = \lambda_Q Q + \lambda_E E - \lambda_C C.$$

where

- Q = quality,
- E = efficiency,
- C = cost.

The controller seeks

$$\max W.$$

17 Warfare Economics

Military organizations allocate resources between:

1. reconnaissance,
2. logistics,
3. command,
4. execution.

The proposed architecture mirrors this structure.

Transformer layers provide command and planning.

SSMs provide logistics and memory.

The controller provides strategic allocation.

18 Algorithms

18.1 Algorithm 1: G-DPSA

Initialize Transformer

Initialize SSM

Initialize Controller

for each batch:

 Compute h_T

 Compute h_S

 Compute breadth B_t

```

    Compute depth  $D_t$ 

     $u_t = R_{\phi}(B_t, D_t)$ 

    Fuse  $h_T$  and  $h_S$ 

    Update parameters

end

```

18.2 Algorithm 2: Reinforcement-Controlled Compute

```

Observe state  $s_t$ 

Select compute budget

Select memory budget

Select routing policy

Receive reward

Update controller

```

19 Future Research

Potential extensions include:

1. diffusion models,
2. robotics,
3. multi-agent systems,
4. economic planning,
5. autonomous scientific discovery,
6. strategic simulations,
7. world models,
8. self-improving foundation models.

20 Conclusion

The Ghoshian Dual Parallel State–Attention Architecture unifies Transformers, State Space Models, Ghoshian Learning, reinforcement learning, hierarchical memory, and expert specialization into a single adaptive framework.

The resulting architecture behaves as a computational economy governed by an executive controller that dynamically allocates learning, memory, reasoning, and compute resources according to breadth and depth signals. This provides a plausible pathway toward more efficient, more autonomous, and more cognitively structured foundation models.

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Glossary

Transformer Attention-based reasoning subsystem.

State Space Model Long-memory linear-time subsystem.

Ghoshian Learning Meta-learning framework that learns optimization dynamics.

Breadth Measure of exploratory diversity.

Depth Measure of convergence and refinement.

Ghoshian Controller Executive control mechanism producing adaptive policies.

Attention Budget Learned allocation of attention resources.

Memory Budget Learned allocation of memory resources.

Mixture of Experts Sparse specialization architecture.

Hierarchical Ghoshian Learning Layer-wise adaptive optimization.

Ghoshian Reinforcement Learning Policy-based control of optimization and compute.

Computational Economy Interpretation of neural computation as resource allocation.

Welfare Function Objective balancing quality, efficiency, and cost.

G-DPSA Ghoshian Dual Parallel State–Attention Architecture.

The End